

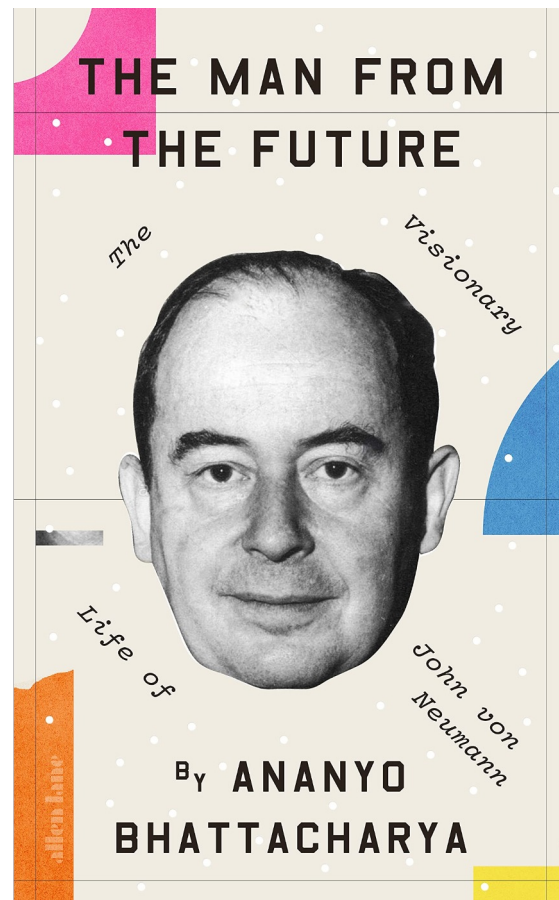
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degli Studi Sociali Guido Carli

von Neumann Architecture

Introduction to Computer Programming

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LUISS



From bombs to computers

April 1944 Los Alamos (base of the Manhattan Project)

Two of the greatest minds minds, Feynman and von Neumann, spend two weeks programming a primitive computer.

The only job of this computer was to solve the integration of a hyperbolic partial differential equation needed for the atomic bomb.

“I am thinking about something much more important than bombs; I am thinking about computers.”



ENIAC

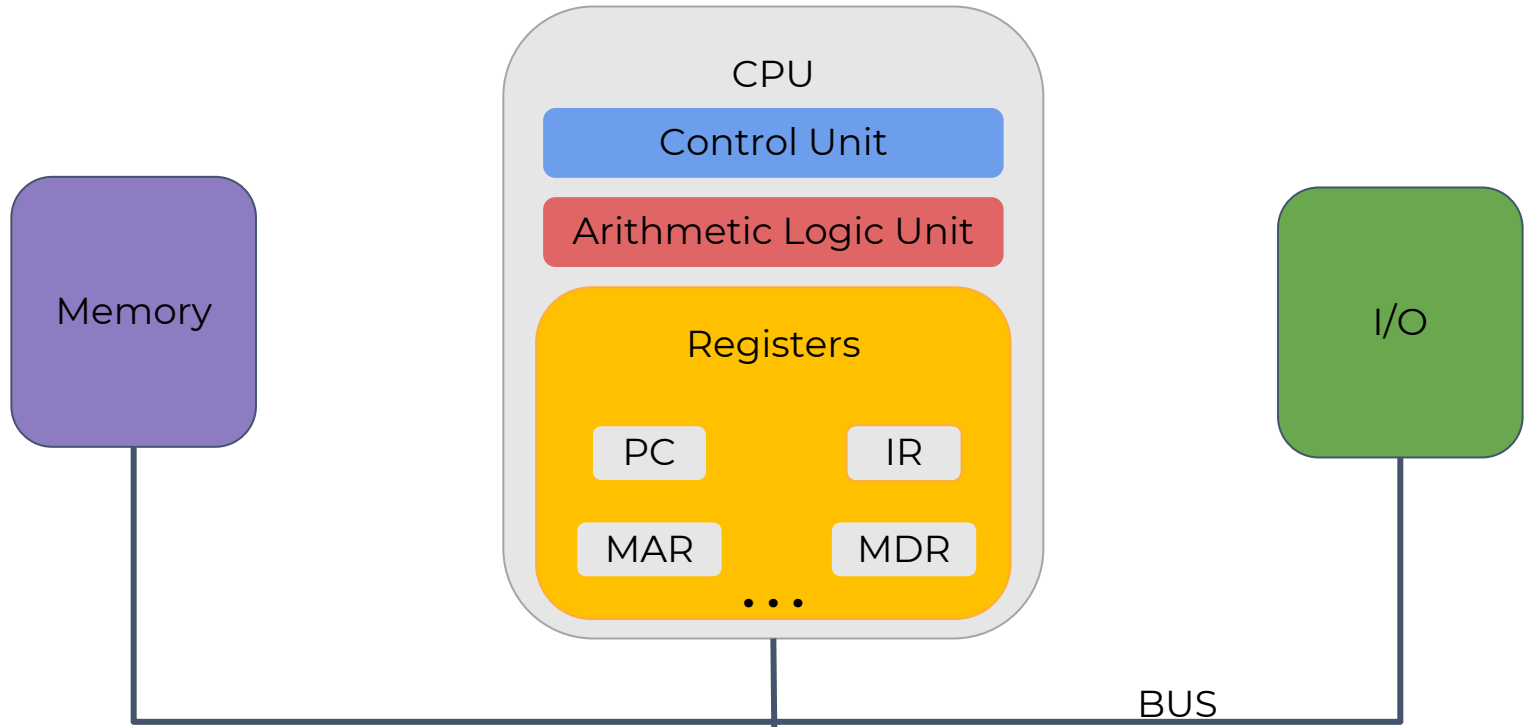
ENIAC (1945) first programmable, electronic, general-purpose digital computer.

It was able to solve a variety of numerical problems through reprogramming (rebuilding the machine 3000 switches, 728 knobs).

The core programming team was formed by 6 women that mapped the problems into ENIAC configurations.



von Neumann Architecture



von Neumann Architecture

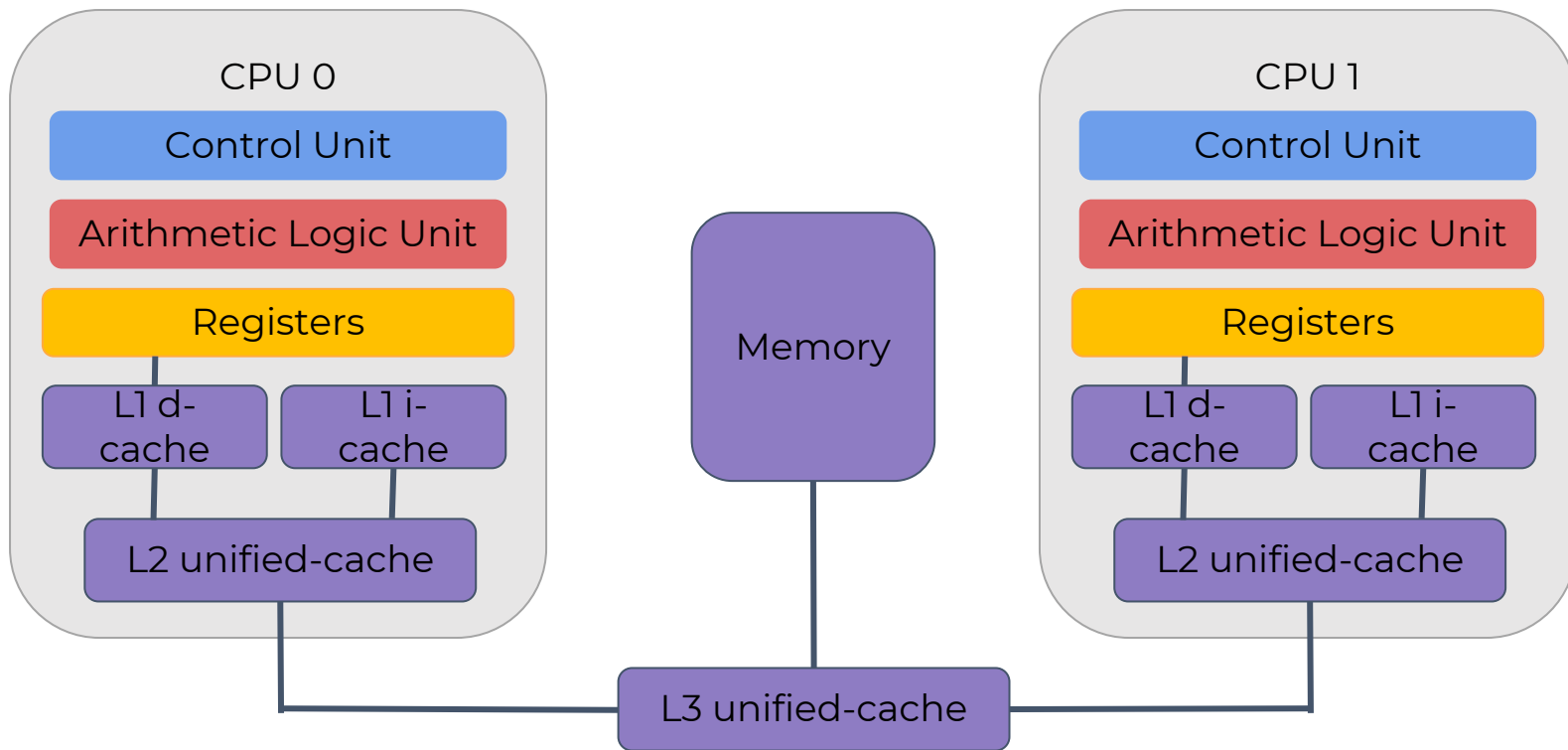
Access one of the classes of memory at a time (instruction and data). The CPU is continually forced to wait for needed data to move to or from memory.

Most modern computers use an evolution of this architecture that alleviates this limited throughput (data transfer rate).

- caches \$\$ (fast and small memories) close to the CPU.
- separated caches for program instructions and data.
- multiple CPU.

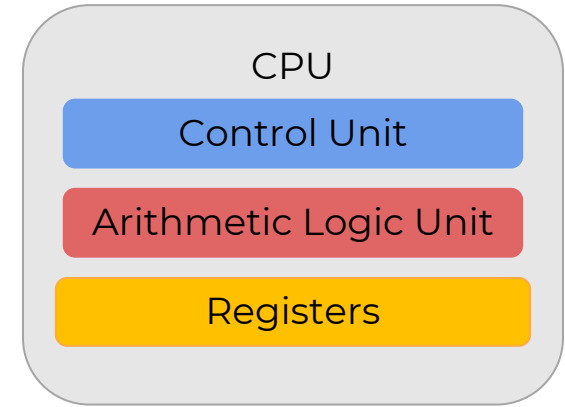


Modern Architecture



CPU: the Brain of the computer

Executes program stored in main memory:



- Control Unit: Fetch and examine instruction.
- Arithmetic Logic Unit (ALU): Performs elementary operations needed to carry out instructions.
- Registers: Small high-speed memory to store temporary results and certain control information.

Control Unit

Coordinates the input and output devices of a computer system.

Coordinates the fetching of program code from main memory to the CPU.

Directs the operation of the other processor components by providing timing and control signals.

Directs the flow of data between the CPU and the other devices.



ALU

The Arithmetic Logic Unit (ALU) is a fundamental component of the Central Processing Unit (CPU) in a computer. It is responsible for performing arithmetic operations (such as addition and subtraction) and logical operations (like AND, OR, NOT) on binary data.

The ALU receives inputs from the instruction decoder and processes them to produce output results, which are then either stored in memory or used in subsequent operations.



Registers

Registers are a small high-speed memory units of the same size inside the CPU.

All registers have the same size.

There are general purpose registers and registers for special use.



Abstraction Levels

A computer uses a hierarchical structure to hide all the complexities and the implementation details to the final user.

Each level is implemented on top of the next lower level and serves as a layer of abstraction.

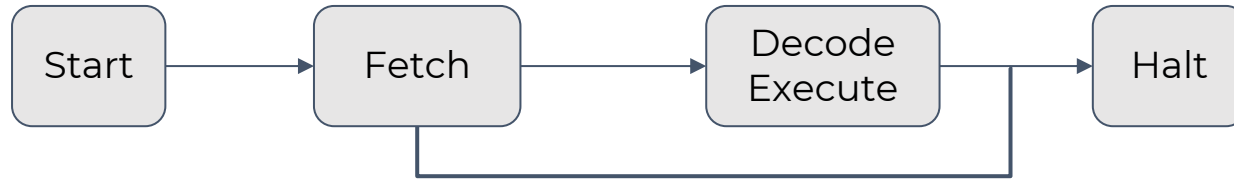
It hides the details of the lower layers from the upper layers.



Fetch Decode Execute

Each instruction is executed in an Fetch-Decode/Execute cycle.

This cycle is regulated by the processor clock.



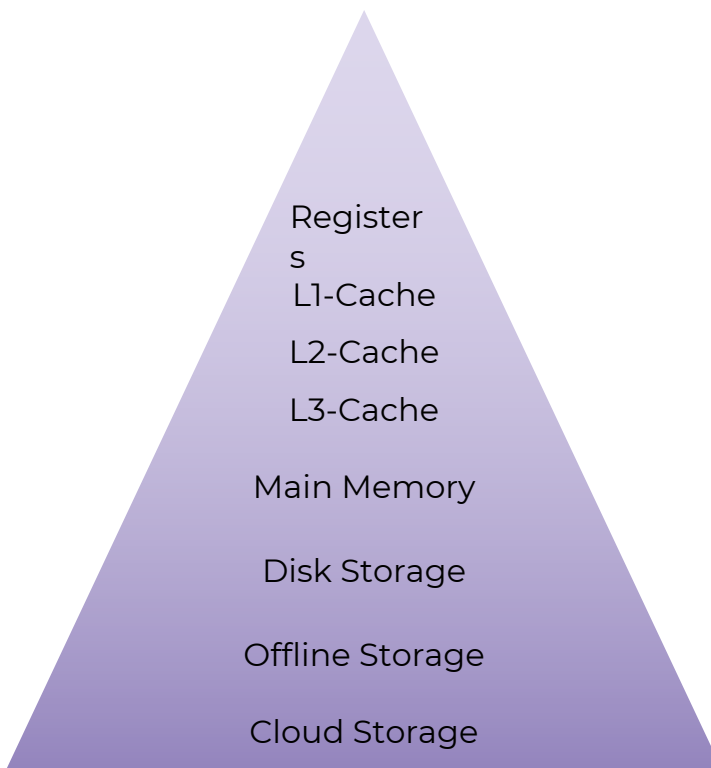
Interrupts

When one instruction has been executed, the fetch-decode/execute cycle moves to the next address.

The processor also checks for interrupts (signals from I/O devices or other sources seeking immediate attention) and either branches to the relevant interrupt service routine or starts the cycle again.



Memory Hierarchy



A purple pyramid representing the memory hierarchy, with levels listed from top to bottom: Registers, L1-Cache, L2-Cache, L3-Cache, Main Memory, Disk Storage, Offline Storage, and Cloud Storage. The pyramid is positioned on the left side of the slide, with its base resting on a decorative blue and dark blue striped bar at the bottom.

Registers
L1-Cache
L2-Cache
L3-Cache
Main Memory
Disk Storage
Offline Storage
Cloud Storage

↑
Speed

↑
Cost

↓
Capacity

Random Access Memory

The Random Access Memory (RAM) is a type of computer memory that is used to store data and machine code currently being used and processed. It is volatile, meaning that the information stored is lost once the power is turned off.

RAM allows data to be read or written in almost the same amount of time irrespective of the physical location of data inside the memory, which enables fast and efficient access to data.

It serves as a computer's "working memory," providing the space for the system to read and write data to be accessed by the CPU. Its capacity often directly influences the speed and performance of a computer system.



Read Only Memory

Data can only be read from the ROM but is maintained when the power is off.

Solid State Drive (SSD): interconnected flash-memory chips. ~5-10x slower Main Memory.

Hard disk drive (HDD): traditional storage device that uses mechanical platters and a moving read/write head to access data. ~20x slower SSD, ~80x slower Main Memory.

Off-line Storage: (SSD, HDD, flash drives) needs the intervention of a user that connects them to the computer. Slower of storage disk due to the connection interface.

Cloud Storage: Someone else computer.



Bus

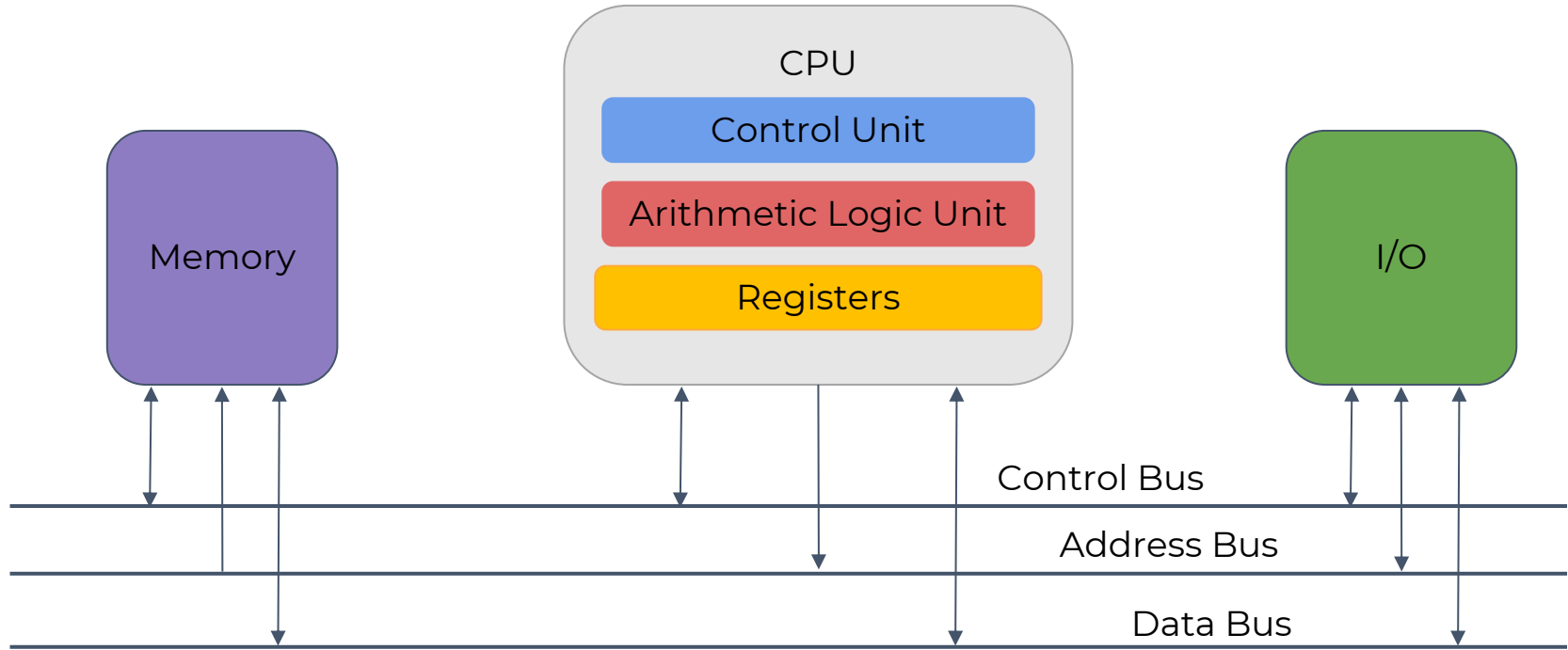
A Bus is a communication pathway.

CPU decides who can read/write on the bus. Everybody else monitor the communications.

A bus consist of multiple channels along which signals, instructions and data will be carried.



System Bus



Operative System

A program that acts as an intermediary between a user of a computer and the computer hardware.

